Advancing Pharmaceutical Distribution Through Data-Driven Inventory Optimization, Demand Forecasting, and Retailer Performance Analytics

Final Report for the BDM Capstone Project

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Contents

Executive Summary:-

Sevam Pharma LLP is a pharmaceutical distribution firm based in Sangli, Maharashtra, operating in the B2B segment and supplying medicines—primarily sourced from Sun Pharma—to retail medical stores across its surrounding region. As the business expanded in terms of product range and retailer network, the company began facing several operational challenges related to inventory and demand management. Slow-moving medicines tend to remain in stock for extended periods, increasing the risk of expiry and financial loss, while fast-moving products often go out of stock, resulting in missed sales opportunities. Additionally, inefficient allocation of inventory across retailers and inconsistencies in stock and sales records have made it difficult to accurately monitor product movement and demand patterns.

To address these challenges, one year of data (February 2025 to January 2026), including product-wise sales, purchase, and stock datasets was collected. The data was consolidated, cleaned, and analyzed using tools such as Python and Excel. Various analytical techniques, including ABC analysis, FSN analysis, inventory comparison, expiry risk assessment, turnover analysis, retailer-level analysis, trend analysis, correlation analysis, and anomaly detection, were applied to evaluate different aspects of business performance.

The analysis enabled the identification of key patterns and trends across sales, inventory, and procurement processes, providing a deeper understanding of product performance, demand behavior, and inventory movement. These insights supported the development of a structured and data-driven approach to decision-making in areas such as procurement planning, inventory allocation, and performance monitoring, contributing to improved operational efficiency and overall business performance.


SEVAM PHARMA LLP
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Detailed Explanation of Analysis Process/Method:-

The organization provided one year of data from February 2025 to January 2026, consisting of three datasets for each month: product-wise sales data, product-wise purchase data, and monthly stock statements. For effective analysis, these monthly datasets were consolidated into unified annual datasets. Sales and purchase data from all months were merged to create a combined sales-purchase dataset, while stock records were integrated into a single comprehensive stock dataset, enabling a holistic view of business operations across the entire period.

The data cleaning process involved examining the structure and format of the datasets to ensure they were suitable for analysis. This included identifying relevant columns, removing redundant or non-informative fields, and checking for missing values. Certain columns, such as discount fields in the sales data and plant-related fields in the purchase data, contained constant values across all records and were therefore excluded as they did not contribute meaningful insights. Additionally, duplicate information such as batch numbers and expiry dates present across datasets was streamlined to reduce redundancy. Missing values in important identifiers like Material Code were handled carefully to maintain consistency and avoid errors in further analysis.

After completing the data cleaning and preprocessing stage, the monthly datasets were combined to create a consolidated annual dataset. This enabled analysis of the pharmacy's performance over the entire period rather than examining each month in isolation. The analysis was carried out using Python, primarily with libraries such as Pandas for data manipulation and Matplotlib and Seaborn for generating visualizations.

For the problem of improving warehouse inventory and expiry manangement using historical stock and sales data, following analytical techniques were applied:

- ***ABC Analysis:*** It is a value-based inventory classification technique used to categorize products based on their contribution to total revenue. In this method, products are arranged in descending order of their revenue contribution and classified into three categories—A, B, and C—based on cumulative percentage contribution. Category A typically consists of a small proportion of products (around 20%) contributing to a large share of revenue (around 70–80%), Category B includes products with moderate contribution, and Category C consists of a large number of products contributing minimally.

The primary objective of ABC analysis is to identify high-impact products that significantly influence overall revenue and require closer monitoring. This helps in prioritizing inventory decisions, ensuring that critical products are always available, while avoiding unnecessary focus on low-impact items.

- ***FSN Analysis:*** FSN (Fast, Slow, Non-moving) analysis is an inventory classification technique used to categorize products based on their movement or consumption rate. In this method, items are classified into three categories: Fast-moving (frequently sold items), Slow-moving (occasionally sold items), and Non-moving (rarely or not sold items) over a defined time period.

  FSN analysis helps in understanding product movement patterns and identify items that may lead to inventory inefficiencies. Fast-moving items require regular replenishment to avoid stockouts, while slow-moving and non-moving items highlight potential overstocking and expiry risks. This helps in improving inventory control, reducing holding costs, and ensuring better alignment between stock levels and actual demand.

- ***Inventory Comparison Analysis (Stock vs Sales):*** It focuses on identifying mismatches between inventory availability and actual demand at a product level. In this method, stock levels are directly compared with corresponding sales quantities over a defined period to evaluate whether inventory is proportionately distributed.

  This analysis focuses on identifying mismatch cases where inventory allocation is not aligned with demand patterns. Unlike movement-based classification, this approach highlights product-level imbalances, such as excess stock concentrated in low-demand items or insufficient stock for high-demand products. This helps in diagnosing allocation inefficiencies and improving decision-making related to procurement and distribution.

- ***Expiry Risk Analysis:*** It is used to identify and evaluate products that are at risk of becoming unsellable due to approaching or past expiry dates. In this method, the expiry dates of products are analyzed alongside their current stock levels to determine whether the available inventory can be sold before expiration. The analysis is performed by comparing stock coverage with the remaining time to expiry using a **risk score** defined as:

  Months of Stock ÷ Months to Expiry

  where *Months of Stock represents how long the current inventory will last based on the rate of sales*, and *Months to Expiry represents the time remaining before the product expires*.

  This analysis helps in quantifying potential financial loss associated with unsold or slow-moving inventory nearing expiry. By linking expiry timelines with sales velocity and stock levels, it highlights specific products that may require immediate action, such as faster liquidation, redistribution, or controlled procurement, to minimize losses.

- ***Inventory Turnover Analysis:*** It measures how efficiently inventory is being utilized by comparing the rate at which products are sold relative to the stock held over a period of time. The analysis is performed using a turnover ratio defined as:

  Sales ÷ Average Inventory

  where *Sales represents the demand or movement of products*, and *Average Inventory represents the typical stock level maintained during the period*.

  This analysis helps in assessing the efficiency of inventory management. A higher turnover ratio indicates that products are sold quickly and inventory is well-utilized, while a lower ratio suggests that stock is moving slowly and may be tied up in the warehouse. This helps in identifying inefficiencies in stock holding and supports better planning of procurement and inventory levels.

For the problem of analyzing the reasons for poor performance of specific retail outlets, following analytical techniques were applied:

- ***Retailer-wise Sales Analysis***: It evaluates the contribution of individual retailers to total sales by aggregating sales value or quantity over a defined period, helping identify key contributors and underperforming outlets. This provides insights into how demand is distributed across retailers and supports better decision-making in inventory allocation and sales planning by prioritizing high-performing retailers while identifying those that may require further attention.

- ***Contribution Analysis:*** It evaluates how much each entity, such as a product or retailer, contributes to overall revenue by expressing individual contributions as a percentage of the total. This helps in identifying major contributors driving business performance and understanding the concentration of sales.

- ***Retailer Return Impact Analysis:*** It evaluates the extent to which returned quantities impact overall sales performance by comparing total supplied stock with the portion returned by each retailer over a defined period. This helps in identifying inefficiencies such as high return volumes, indicating misalignment between supplied inventory and actual demand.

For the problem of identifying seasonal and regional demand patterns, following analytical techniques were applied:

- ***Revenue Trend Analysis:*** It examines how overall revenue performance changes over time by analyzing revenue across different periods such as months. This helps identify patterns such as growth, decline, or fluctuations, providing a macro-level view of business performance. It is useful for understanding how the business evolves over time and for identifying periods of high or low revenue activity.

- ***Demand Pattern Analysis (time-based):*** It focuses on understanding variations in demand at a granular level by identifying recurring patterns such as seasonality or periodic fluctuations in product consumption. By analyzing these patterns over time, it helps in recognizing predictable demand cycles and shifts in consumption behavior. This supports more accurate forecasting and better alignment of procurement and inventory decisions with expected demand variations.

For the problem of understanding the causes of anomalies and inconsistencies in stock and sales data, following analytical analysis were done:

- ***Correlation Analysis:*** It is used to examine the relationship between different variables by measuring how changes in one variable are associated with changes in another. It is typically represented using a correlation coefficient that ranges from -1 to 1, where values close to +1 indicate a strong positive relationship, values close to -1 indicate a strong negative relationship, and values near 0 indicate little or no relationship.

  This analysis helps in identifying whether key business variables, such as sales, revenue, profit, and purchases, are aligned with each other. It is particularly useful for detecting inconsistencies or unexpected relationships, such as situations where higher purchases do not correspond to higher sales, thereby highlighting potential inefficiencies in operations or decision-making.

- ***Anomaly Detection Analysis:*** It is used to identify unusual or unexpected patterns in the data that deviate significantly from normal behavior. In this context, it involves examining variables such as sales, purchases, and stock levels to detect outliers or inconsistencies that do not follow typical trends.

  This analysis helps in uncovering potential data errors, operational irregularities, or abnormal business activities, such as sudden spikes or drops in sales, mismatches in stock records, or unusually high purchase values. Identifying such anomalies is important for ensuring data reliability and for investigating underlying issues that may impact business performance.

Results and Findings:-

The analysis of the consolidated dataset revealed several important patterns, trends, and inconsistencies in sales, procurement, and inventory management. The following results highlight key observations derived from the data and their implications for business performance, organized based on the applied analytical methods.

- ***ABC Analysis:-***

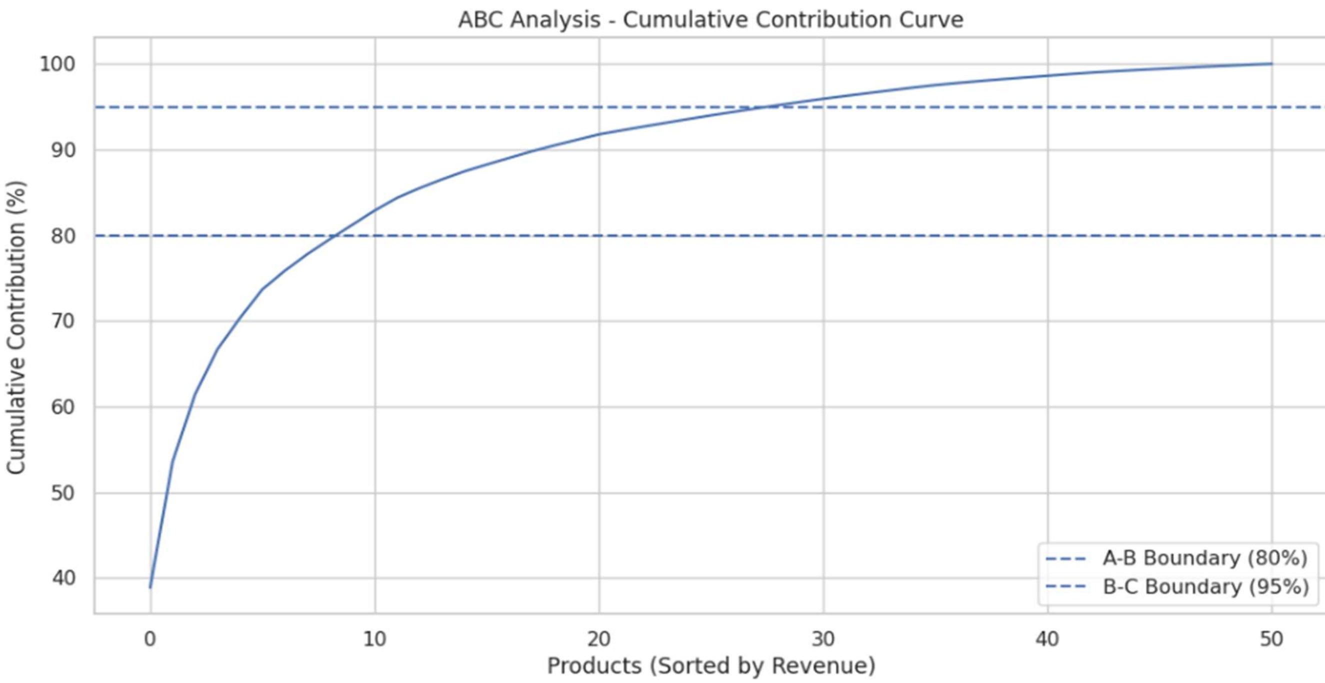

Figure 1: Cumulative Contribution Curve for ABC Analysis

The ABC analysis shows a clear concentration of revenue among a small number of products. Category A consists of **9 products contributing to approximately 80% of total revenue**, indicating strong dependence on a few high-value items, with the top product **(PRAMIPEX ER 0.375 TA 10 TAB)** alone contributing around **38.85%**. Category B includes **19 products contributing the next 15%**, while Category C comprises **23 products contributing only about 5%** , representing a long tail of low-value items. The cumulative contribution curve reflects a steep initial rise followed by a gradual flattening, confirming a Pareto-like distribution where a small proportion of products drives the majority of revenue.

- ***FSN Analysis:-***

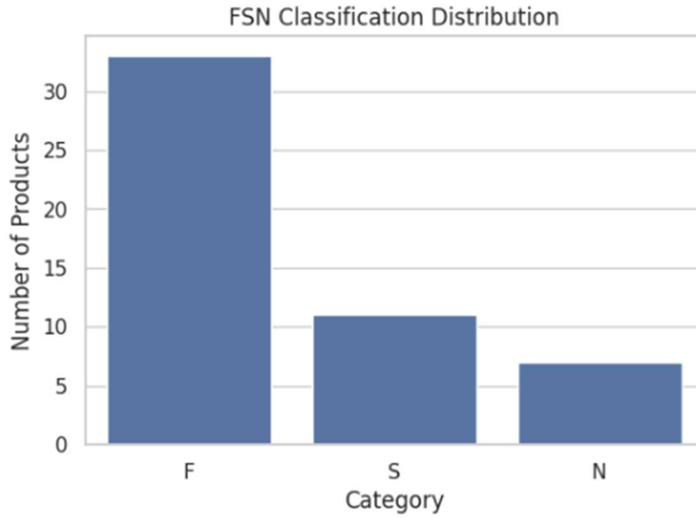

Figure 2: Bar Plot for FSN Analysis

The FSN analysis shows that a majority of products fall under the Fast-moving category, with **33 products classified as F**, including **TRAPEX 2MG TAB 10 T** and **QUTIPIN SR 50 TAB 10 TAB**, indicating consistent demand. The Slow-moving category consists of **11 products**, such as **QUTIPIN 200 MG TAB 10 T** and **LEVIPIL INJ. 5 ML**, while only **7 products fall under the Non-moving category**, including **PRIMOX 25MG TAB 10TAB** and **ADMENTA 10 MG TAB 10 TAB**, reflecting limited sales activity.

- ***Inventory Comparison Analysis (Stock vs Sales):-***

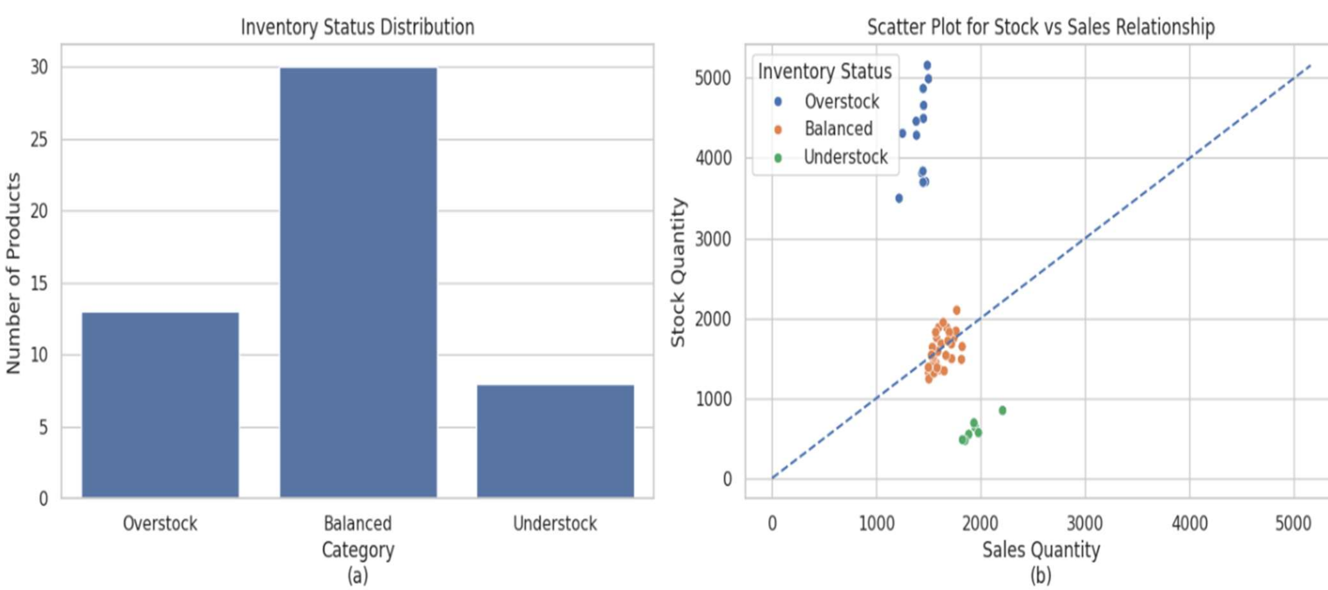

Figure 3: Visualizations of Stock vs Sales

The inventory comparison analysis shows that **30 products are balanced**, **13 are overstocked**, and **8 are understocked**, indicating that while most products have stock levels aligned with demand, a significant portion still exhibits imbalance. Overstocked products reflect excess inventory and potential holding or expiry risks, whereas understocked products indicate possible stockouts and missed sales opportunities. The scatter plot further highlights this pattern, with overstocked items lying above and understocked items below the ideal stock–sales line.

Top overstock products: MAXDULIN 50/20 CAP 10T, PALIRIS LA 150 MG, QUTIPIN 25 MG TAB 10 T, ARAVON I V 20 ML, TOFICALM 50 10TAB
Top understock products: QUTIPIN 50 TAB 10 TAB, LEVIPIL RTU 500 MG 100ML, VORTIDIF 10 MG TAB 10, QUTIPIN SR 50 TAB 10 TAB, MIRTAZ 7.5MG TAB 10 T

- ***Expiry Risk Analysis:-***

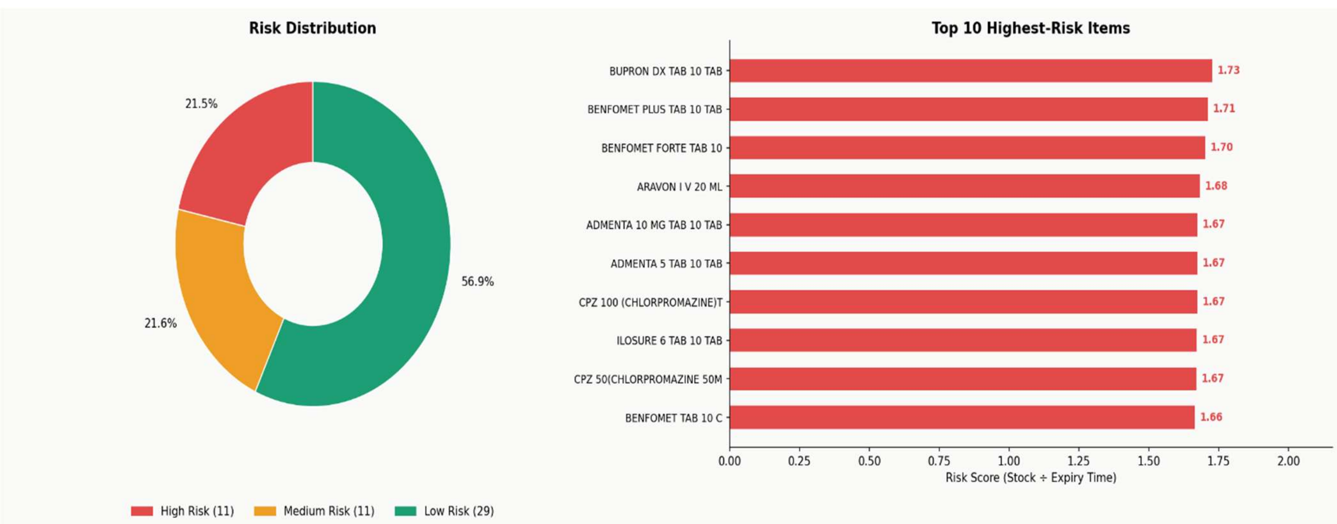

Figure 4:(a)Pie Chart of Products based on risks (b) Bar Plot of Product vs Risk Weight

Most of the inventory (**56.9%**) falls under the low-risk category (**score < 0.75**), indicating that a majority of products have sufficient time to be sold before expiry. However, **21.5%** of products are high risk (**score > 1**), showing a significant portion where stock levels exceed the time available for sale. Another **21.6%** lie in the medium-risk (**score ≈ 0.75 – 1**) range, requiring monitoring. The top high-risk items exhibit higher stock relative to expiry time, highlighting potential slow-moving inventory that may lead to losses if not managed properly.

- ***Inventory Turnover Analysis:-***

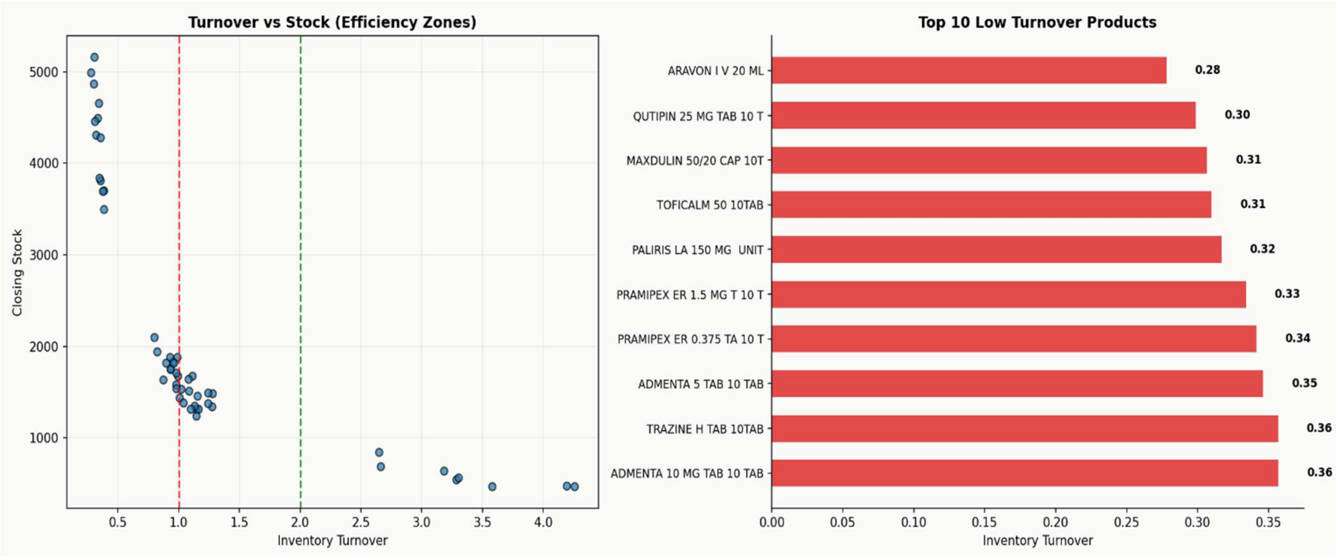

Figure 5:(a) Scatter Plot of Products (b)Bar Plot of Product vs Inventory Turnover(Top 10 Inefficient)

The turnover analysis indicates that a significant portion of inventory falls under low efficiency (≤ **1**), reflecting slow-moving stock and potential overstock situations. A moderate share lies within the **1–2** range, suggesting relatively balanced movement but with room for improvement. Only a smaller portion achieves high efficiency (> **2**), indicating optimal stock utilization. The scatter plot further shows that several low-efficiency items maintain **high closing stock despite low turnover**, highlighting an imbalance between supply and demand and the need for better inventory optimization.

- ***Retailer-wise Sales and Contribution Analysis:-***

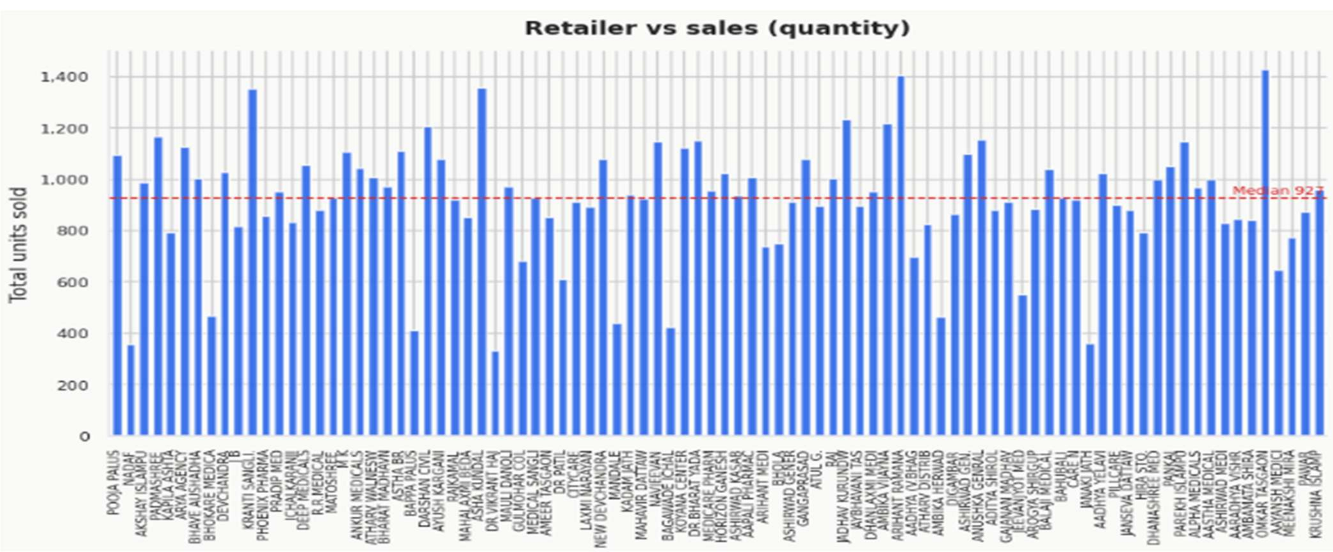

Figure 6:Distrubution of Sales among retailers

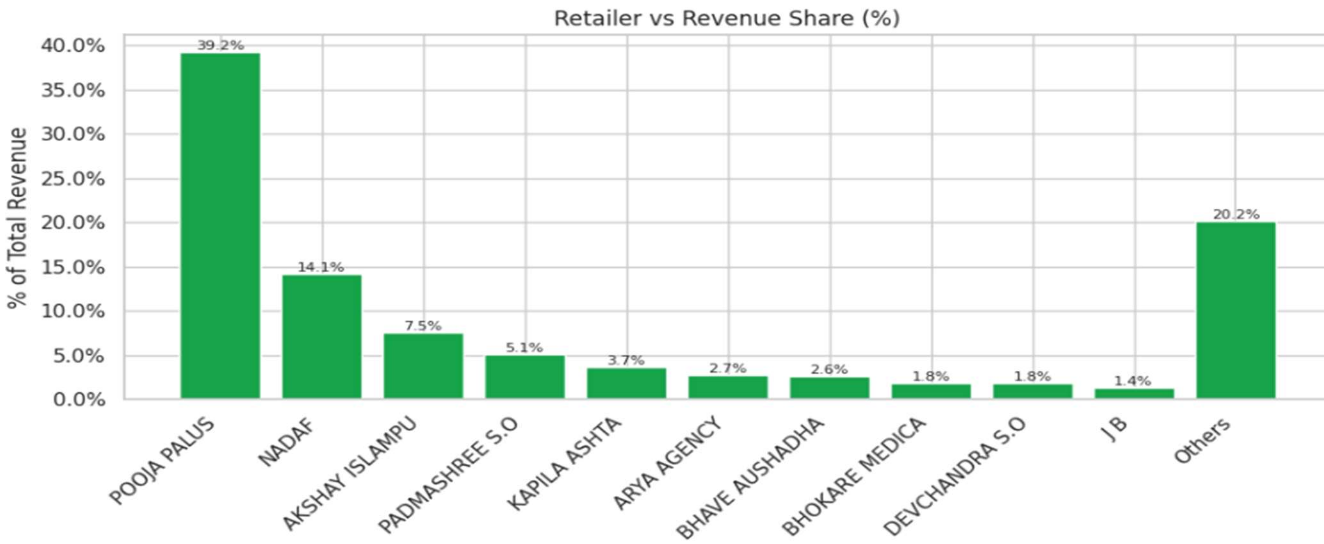

Figure 7: Revenue contribution among retailers

The retailer-wise analysis shows that sales quantity is fairly distributed across most retailers, with values clustering around a similar range, indicating consistent product movement across outlets. However, the revenue distribution is highly concentrated, with a single retailer contributing approximately **39%** of total revenue, followed by a sharp decline among subsequent contributors, while most others contribute relatively small shares and are grouped under "**Others**." It is also observed that retailers such as **NADAF** and **Devchandra**, despite having relatively lower sales quantities, contribute significantly to revenue, indicating a focus on **higher-value medicines** and highlighting differences in product mix across retailers.

- ***Retailer Return Impact Analysis:-***

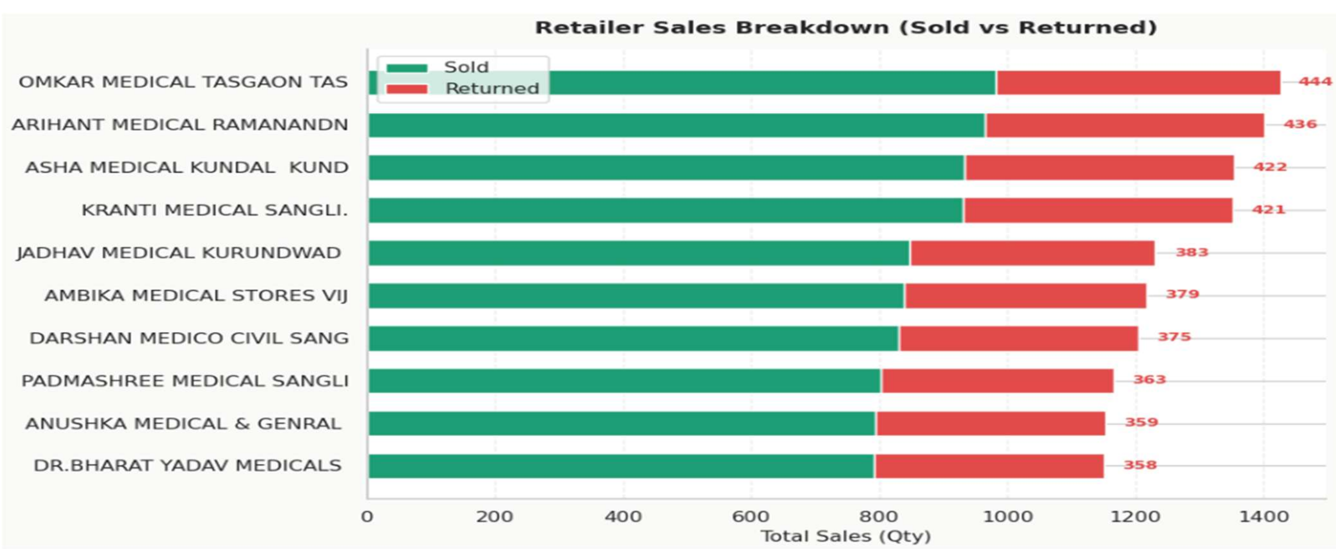

Figure 8: Stacked Bar Plot for Retailer Sales Breakdown (Top 10)

The retailer-level sales breakdown shows that returns constitute a substantial portion of total transactions, with several retailers returning approximately **350 to 450 units** out of total sales ranging between **1,100 and 1,400 units**, indicating that nearly **25–35%** of the supplied quantity is not effectively realized as final sales. This highlights that despite sufficient stock availability, a significant share of goods is being returned, limiting revenue generation. The consistent presence of high return volumes across retailers suggests inefficiencies in stock allocation, where inventory does not fully align with actual demand, ultimately affecting overall sales performance.

- ***Trend Analysis:-***

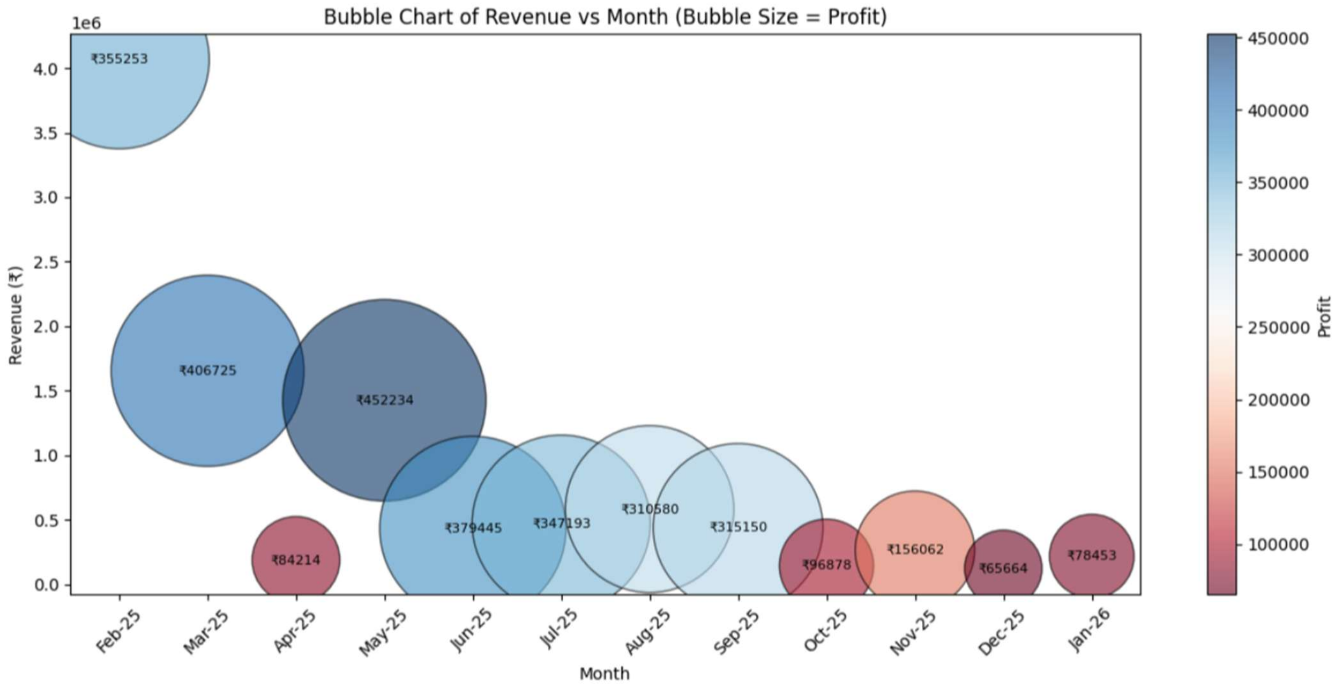

Figure 9: Bubble Chart of Revenue vs Month

Months with higher revenue do not always show larger bubbles, indicating that profit is not directly proportional to sales volume. While **February** records the highest revenue, months like **March** and **May** exhibit larger bubbles, reflecting higher profitability. The noticeably smaller bubbles from **October** to **January** indicate lower profitability during this period, likely because retailers stock more heavily in the earlier months and require only limited replenishment toward the end of the cycle.

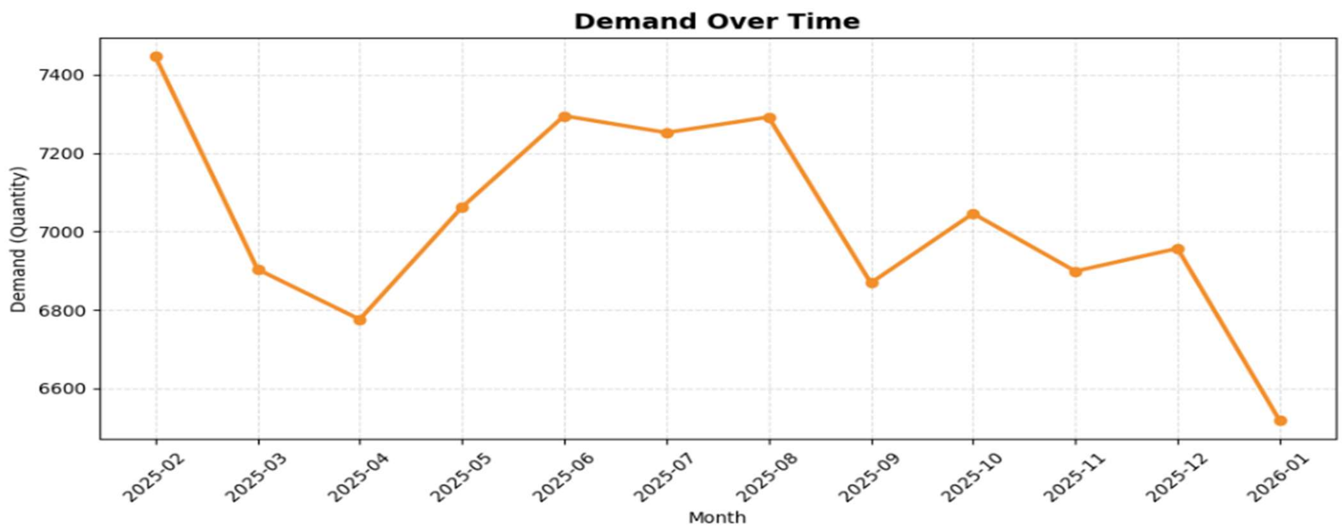

Figure 10: Line Chart of Demand over Month

The demand trend shows a generally stable pattern with moderate fluctuations over time. Demand is highest at the beginning of the period (**February**), followed by a decline in **March** and reaching its lowest point around **April**, which also corresponds to lower profit. It then recovers from **May to August**, indicating improved consumption or restocking activity. A dip is observed again in **September**, where profit remains relatively stable despite the decrease in demand. Demand rises in **October**; however, profit declines during this period, suggesting reduced margins. Thereafter, demand stabilizes with minor fluctuations until **December**. The sharp decline in **January** indicates reduced demand or minimal restocking toward the end of the cycle.

- ***Correlation Analysis:-***

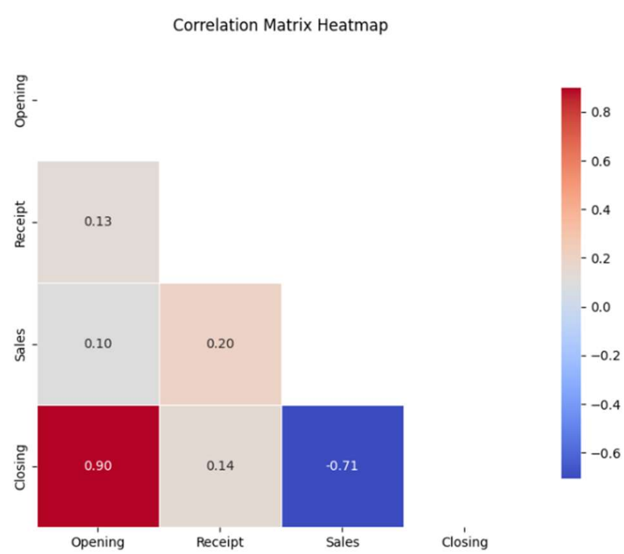

Figure 11: Correlation between the stock

The correlation matrix indicates that opening and closing stock are strongly positively related (**0.90**), showing that inventory levels are largely carried forward with limited variation. Sales exhibit a strong negative correlation with closing stock **(−0.71)**, confirming that higher sales lead to a reduction in remaining inventory. The relationship between sales and receipts is weakly positive (**0.20**), suggesting that replenishment is only partially aligned with sales activity. Additionally, the weak correlations between opening stock and sales (**0.10**) and between receipts and closing stock (**0.14**) indicate limited direct influence, highlighting some inefficiencies in inventory planning and stock allocation.

- ***Anomaly Detection:-***

During anomaly detection, it was observed that certain products have missing or inconsistent material codes across datasets. This creates challenges when merging datasets based on product names, as the same product may be recorded with different naming conventions, leading to mismatches and data inconsistencies.

Additionally, the stock maintenance dataset records opening and closing stock for each month, which should ideally satisfy the condition **Closing Stock = Opening Stock + Purchase – Sales**. However, several records do not follow this relationship, and products with no purchase or sales are entirely missing from the dataset. Such products should still be included with equal opening and closing stock values to ensure completeness. These issues highlight gaps in data recording practices, contributing to inaccurate inventory tracking and overall mismanagement.

- ***Link to Dataset and Code:-***

https://drive.google.com/drive/folders/1bJx9N-o4fXlg1Ojbqj3zNyEu55Xlmnzo?usp=sharing

Intepratation of results and Recommendations:-

The following interpretations can be derived from the observed results and findings:

- A small group of products contributes to the majority of revenue, with one product alone contributing a very high share. This shows that the business is **highly dependent on a few medicines**, making it risky if their demand drops or supply is disrupted.
- Even though most products are classified as fast-moving, there are still several **slow and non-moving items occupying inventory space**, which means money is tied up in products that are not generating returns.
- The presence of both **overstocked and understocked products** clearly indicates that stock is not being allocated based on actual demand. Some products are lying unsold, while others are not available when needed, leading to missed sales.
- The very high correlation between opening and closing stock shows that **inventory is mostly being carried forward month to month**, which means sales are not significantly reducing stock levels. This is a strong sign of overstocking or low movement.
- The weak relationship between sales and receipts suggests that **procurement is not directly linked to actual sales**, meaning products are being purchased without fully considering how much is being sold.
- Retailer analysis shows that while sales quantity is somewhat evenly distributed, **revenue is concentrated among a few retailers**, indicating differences in product mix. High return rates (25–35%) further suggest that **retailers are receiving more stock than they can sell**, leading to inefficiencies.
- Trend analysis shows that **high sales do not always result in high profit**, meaning some products or periods have lower margins. Also, demand fluctuates across months, indicating **seasonal or stocking behavior by retailers**.
- Data inconsistencies, such as missing material codes and incomplete stock records, make it difficult to track products accurately and lead to **errors in analysis and decision-making**.

Based on the above interpretations we can make use of these following recommendation and strategies:

- Adopt a **sales-based replenishment strategy** where purchase quantities are calculated using recent sales trends. For example, maintain stock equal to **1.2–1.5 times the average sales of the last 2–3 months**, instead of bulk purchasing without reference to demand.
- Define clear **reorder levels (minimum stock thresholds)** for each product. When stock falls below this level, trigger replenishment automatically to avoid understocking and missed sales.
- For overstocked products, take targeted corrective actions such as:
    i. Offering **discounts or bundled schemes** to retailers
    ii. **Redistributing excess stock** to retailers with higher demand
    iii. Temporarily **reducing or pausing procurement** of those products
- For slow and non-moving items, implement a **clearance strategy**, including limited-time offers or replacing them with better-performing alternatives to free up inventory space and working capital.
- Shift from uniform distribution to a **retailer-specific allocation model**, where stock is supplied based on each retailer's historical sales performance rather than sending equal quantities to all retailers.
- Reduce high return rates by introducing a **controlled dispatch system**, where maximum supply limits are set for each retailer based on their past consumption to prevent overstocking at the retailer level.
- Move from bulk purchasing to a **frequent small-batch procurement approach**, ensuring that inventory moves faster and reducing the risk of expiry and holding costs.
- Introduce **expiry-based prioritization**, where products nearing expiry are identified early and pushed through targeted promotions or faster distribution channels.
- Focus on **high-margin products** in addition to high-selling ones. Prioritize stocking and promoting products that contribute more to profit rather than just revenue.
- Improve inventory turnover by regularly monitoring **stock-to-sales ratios** and taking corrective action when stock remains unsold beyond a defined time period.
- Standardize **product identification across datasets** by maintaining consistent material codes and naming conventions to avoid mismatches during analysis and reporting.
- Ensure **complete stock recording**, including products with no sales or purchases, by maintaining their opening and closing stock values to improve data accuracy.

- Regularly validate inventory data using the rule: **Closing Stock = Opening Stock + Purchase – Sales**, and flag any violations for correction.
- Implement a **monthly inventory review system**, where key metrics such as stock levels, returns, turnover, and expiry risk are analyzed to take timely decisions.
- Use **data-driven decision-making dashboards** to track product performance, retailer contribution, and stock movement in real time, enabling quicker and more accurate planning.
- Develop a **balanced inventory strategy**, where dependence on a few high-performing products is reduced by improving the performance and visibility of mid- and low-tier products.

Conclusion:-

This report highlights the importance of adopting data-driven strategies to improve inventory and operational efficiency at Sevam Pharma LLP. The analysis revealed key challenges such as misalignment between stock and demand, excess inventory leading to expiry risks, stockouts of high-demand products, high return rates from retailers, and inconsistencies in data management, all of which impact revenue and overall performance. Addressing these issues requires implementing demand-based replenishment, controlled inventory allocation, and regular monitoring of stock movement to ensure better utilization of resources. Improving data accuracy and focusing on high-margin products can further enhance profitability while reducing losses. Overall, strengthening existing operations through better inventory control and planning will provide a stable foundation for improved efficiency, increased revenue, and sustainable business growth.

-Thank you.